

SWARALI PAYGUDE

swaralipaygude@gmail.com | 213-574-5212 | [linkedin.com/in/swaralipaygude](https://www.linkedin.com/in/swaralipaygude)
San Francisco, CA (open to relocation)

SUMMARY

An organized and perseverant individual, eager to utilize her organizational skills and time management across diverse settings. Dedicated and adaptable professional with a strong work ethic, effective communication skills, and the ability to learn quickly. Seeking opportunities to enhance abilities while contributing to a dynamic team, and supporting organizational goals.

EXPERIENCE

Relyance AI

Software Engineer(Associate) | Data Team

San Francisco, CA

MAR 2025-DEC 2025

- Owned and led the complete revamp of the Vendor Management app for QC and Legal teams — driving requirements, feedback cycles, and migration from risky raw SQL operations on production to a secure, API-driven Django architecture.
- Built a clean, responsive UI with extensive client-side validation, faster navigation, and backend constraint enforcement; added critical backend endpoints and wrote migration scripts for 25,000+ vendor records with zero data loss, improving QC team efficiency and throughput by ~2x.
- Created and integrated an AI-powered data enrichment pipeline (Gemini) into the frontend to auto-populate key vendor data, reducing manual research and entry time by 50–60%.
- Developed GitHub Actions CI/CD pipeline to auto-generate schemas and open PRs, cutting manual maintenance time by 80%; implemented comprehensive unit testing for the IAM repository, improving reliability and reducing regression issues by ~40%.

Cylerian

Software Engineer Intern, Detection Engineering

Los Angeles, CA

DEC 2024-MAR 2025

- Executed MITRE ATT&CK-aligned Atomic Red Team tests using PowerShell to assess detection coverage and identify gaps in threat visibility.
- Developed and refined Sigma-based detection rules in CyQL and YAML to detect adversary behaviors and improve alert accuracy.
- Automated atomic test execution using PowerShell scripts, streamlining validation workflows and reducing manual effort.

UpDonor

Software Engineer Intern

Los Angeles, CA

MAY 2024-AUG 2024

- Created a public key system for UpDonor's Enterprise Plan with Nodejs, enabling secure, limited API access for nonprofits, with capacity to handle 300+ requests per minute.
- Designed a key management dashboard in React with features for generating, displaying, and revoking API keys, empowering organizations with quick, easy-to-use, secure access controls.
- Implemented express middleware to verify API key usage, integrated PostgreSQL for secure storage and management of 1000 API keys.

USC Annenberg Inclusion Initiative

Data Engineer | Data Management Team

Los Angeles, CA

SEP 2022-DEC 2023

- Optimized data validation for 35 variables in Google Sheets using JavaScript, utilizing reusable add-on script with Google Apps Script, resulting in saving time spent on continual manual operations.
- Devised data verification algorithm for 25000+ rows of compiled IMDB data against actual data found on IMDB website.

EDUCATION

University of Southern California, Viterbi School of Engineering

Master of Science in Computer Science (3.71/4.00)

Los Angeles, CA

JAN 2022-DEC 2023

MIT World Peace University

Bachelor of Technology in Computer Science and Engineering (9.41/10.00)

Pune, India

AUG 2017-MAY 2021

TECHNICAL SKILLS

- Languages: Python, JavaScript/TypeScript, HTML/CSS, Java, C++, SQL
- Frameworks/Tools: Node.js, React, Django, Flask, PostgreSQL, MongoDB, Git, Docker, Postman, Pandas, Tooljet
- Cloud: AWS, GCP, Azure

ACADEMIC PROJECTS

Event Finder Website | *HTML, CSS, Javascript, Python Flask, REST APIs, JSON, GCP*

- Built responsive website allowing users to easily access information about 50000+ public events, with response time of 500 ms; Hosted on Google Cloud Platform App Engine.

EventExplorer: Advanced Event Search and Details Android App | *Android Studio, Java, Android SDK, NodeJS, XML*

- Constructed end-to-end Android application from scratch, featuring intuitive interface for users to search and access information about various events in a region, streamlining the process by presenting all relevant details with a single click.

Automated PCB Fault Detection System | *Python, Image Processing, Research paper publication*

- Conducted literature survey on fault detection in Printed Circuit Boards; expanded dataset by 5x using image augmentation techniques.
- Crafted algorithm in Python for detecting PCB defects with 79% accuracy.
- Compared Image Processing and Transfer Learning techniques; published findings in IJRASET 2021 ([link to paper](#)).

Word Pursuit: Unity Game | *C#, Unity, WebGL, Scrum, Git*

- Collaborated in a team of 8 to build a word-guessing game combining different genres of arrow shooting and hangman. Utilized C# on Unity platform with Git for version control, launched live gameplay on WebGL.
- Engineered game mechanics, collected gameplay data from 100+ users through testing for performing data analytics, improved user experience based on feedback analysis.